

Shavin Fernando

Senior Software Engineer

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PROFILE

IFS Cloud Developer with 3+ years of hands-on experience at IFS R&D International, working across all layers of the IFS Cloud stack - Aurena client development (.client/.fragment), projection framework (.projection), server-side logic (.plsvc/.plsqli), Java EE server extensions, database schema design, and IFS Report Studio Designer. Delivered end-to-end features across 5 IFS components (CRM, RMCOM, RMPANL, Order, HCM) spanning modifications, reports, configurations, third-party integration (Microsoft Teams), and AI/ML platform integration (IFS Copilot, LLM recipes). Experienced with EBR implementations, data pumping, upgrade scripts, and IFS Developer Studio. Previously worked in the HCM module during internship, gaining proficiency in the IFS Marble framework. Visiting Lecturer at IIT teaching Database Systems, Java, and client-server architecture. Combines deep IFS product knowledge with strong PL/SQL and Java fundamentals, a product-ownership mindset, and a track record of mentoring and cross-timezone collaboration.

TECHNICAL SKILLS

Languages PL/SQL, Java, SQL, Go, Python, TypeScript, JavaScript	IFS Cloud Development Aurena (.client, .fragment), Projections, Server-side (.plsvc, .plsqli, .rdf, LU development)	IFS Reports IFS Report Studio Designer, report models (.report), RPT/REP/RPK artifacts, Report_Format_API, PDF generation pipeline	IFS Tools & Practices IFS Developer Studio, EBR (Edition-Based Redefinition), data pumping, upgrade scripts (.upg), DDL/storage definitions, .ins scripts
Java Ecosystem Java EE / Jakarta EE, EJB, JDBC, JAX-RS, AES Cryptography (javax.crypto)	Databases Oracle (PL/SQL, JSON_TABLE, JSON_ARRAYAGG), PostgreSQL, Prisma ORM, Redis, Database Schema Design, Performance Tuning	Integration Microsoft Teams Platform, OAuth 2.0, IFS ML/LLM Platform, REST APIs, Cross-module	DevOps & Cloud Docker, Kubernetes (Lens), GitHub Actions CI/CD, Git, Bitbucket, Jira, Agile/Scrum
Testing & Quality PL/SQL Unit Testing, API Test Automation (TAR), Postman, Code Review			

PROFESSIONAL EXPERIENCE

CRM, Service & Asset Management, IFS R&D International (Pvt) Ltd, Sri Lanka, Software Engineer

June 2024 – Present

IFS Aurena Development & Customizations

- Developed Aurena client pages (.client) and reusable fragments (.fragment) across CRM, RMCOM, RMPANL, and Order components - building new pages from scratch (Guided Next Steps, Settings, Recommendation Deletion) and extending existing ones (Business Opportunity, CRM Account, Sales Quotation, Customer Agreement).
- Implemented UI features including LOV selectors, multi-edit support, dialog workflows, assistant wizards (AI subscription), badge emphasis for states, navigator entries, and confirm messages with validation logic.
- Designed and implemented IFS Cloud projections defining entity overrides, CRUD endpoints, custom actions, functions, and queries - consumed by Aurena clients and external integrations.
- Built projection service logic (.plsvc) implementing complex business workflows: state machine transitions, business object lifecycle management, and cross-module data retrieval.

PL/SQL Business Logic & Database Design

- Developed PL/SQL business logic across Logical Units including creating entire LUs from scratch with full lifecycle management, validation, and cascading operations.
- Designed relational database schemas including new tables, upgrade scripts (.upg), DDL definitions (.ddlsource), storage configurations (.storage), and custom views (.views) across CRM and RMCOM components.
- Implemented complex state machine workflows for Business Opportunity Revisions with cascading line status updates, agreement state synchronization, and cross-module order status updates.
- Built advanced JSON processing pipelines using Oracle JSON_TABLE, JSON_ARRAYAGG, and JSON_OBJECT for real-time data aggregation from multiple CRM entities for AI/ML consumption.

IFS Report Development

- Designed and built the Business Opportunity PDF report from scratch using IFS Report Studio Designer, featuring a hierarchical 13-block data model with ~90 columns covering opportunity headers, revisions, line items, contacts, accounts, representatives, competitors, and economic summaries.
- Implemented the full report generation pipeline: projection action (GetResultKey) -> plsvc function -> Report_Format_API.Create_New_Report -> RPI package execution -> PDF output via print dialog.
- Created the client-side "Print Opportunity" command integrating the report into the existing Business Opportunity Aurena page workflow.

Integration (Third-Party & Cross-Module)

- Built end-to-end Microsoft Teams CRM integration: OAuth client secret encryption (Java EE/AES), dynamic Teams app manifest ZIP generation, meeting participant context APIs, AI-powered meeting briefing (LLM recipe + PL/SQL JSON processing), and meeting summary persistence as business activities.
- Implemented cross-module integration between CRM and Order components, linking Business Opportunities with Customer Agreements, Sales Quotations, and Customer Orders via shared projection references.

- Integrated with the IFS ML/LLM Platform: authored GPT-4o prompt recipes (.ins) for meeting summarization, guided next step recommendations, and AI Copilot customer evaluations. Consumed ML APIs via ML_Model_API and ML_SYS.
- Extended the IFS AI Copilot (Sales Representative Assistant) with a Customer Evaluation capability - built the evaluation data layer (schema, PL/SQL, prompt recipes, service projection) end-to-end.

Configuration & EBR

- Built configuration pages for feature management (GuidedNextStepSetting) with admin-facing Aurena UI and projection APIs, enabling per-feature toggle controls for AI recommendation generators.
- Learned and documented EBR (Edition-Based Redefinition) implementation patterns for the CRM team, delivered sessions to engineers, and created an automated EBR PR Review agent (Rovo) on Bitbucket.
- Created and maintained data pumping scripts, documented processes on Confluence, and delivered KT sessions on data pumping strategies.

Testing, Leadership & Collaboration

- Wrote PL/SQL unit tests and TAR (test automation) scripts for regression coverage. Performed manual testing across features before deployment.
- Delivered KT sessions and technical presentations on AI Copilot, performance optimization, data pumping, and EBR implementations to 30-50 engineers across cross-functional teams spanning the US, Germany, and Sweden.
- Mentored a new intern and participated in intern recruitment interviews.
- Contributing to IFS's next-generation Kairos platform built on Go (backend) and React (frontend).

Informatics Institute of Technology, Visiting Lecturer September 2024 – Present

- Teaching 3-4 modules concurrently to 160+ students (1st year to final year): Database Systems, Python, Java & OOP, Web Development, and Client-Server Architecture (JAX-RS, Jakarta EE, server sockets, concurrency, stream I/O).
- Conducting 4-6 classes per week, evaluating final-year dissertations, and running viva assessments.
- Mentoring final-year students on their projects and conducting mock interviews for career preparation.

Human Capital Management, IFS R&D International (Pvt) Ltd, Sri Lanka, July 2022 – September 2023

Intern - Software Engineer

- Developed IFS Cloud application features using PL/SQL and IFS frameworks within the HCM module, contributing to the Human Capital Management product used by enterprise customers globally.
- Implemented API-based test automation scripts and PL/SQL unit tests for regression coverage across HCM components.
- Built proficiency in PL/SQL and the IFS Marble framework, designing entity-relationship models and implementing business logic within IFS Cloud's enterprise architecture.
- Gained hands-on experience with Kubernetes pod management using Lens, including environment restarts and container-based secret retrieval for development workflows.
- Worked within Agile methodologies using Jira, Bitbucket, and SourceTree for sprint planning, code reviews, and delivery.

OTHER PROJECTS

Multi-Tenant E-Commerce SaaS Platform, *Next.js 15, React, TypeScript, PostgreSQL, Prisma, Stripe, Docker, GitHub Actions*
Architected a white-label e-commerce platform as a Turborepo monorepo with ORM-level tenant isolation, Docker Compose local environment, GitHub Actions CI/CD pipeline, and cloud infrastructure (Neon, Upstash, Cloudflare R2, Vercel).

CryptoVisionX, *Python, ResNet50, LSTM, CNN, Grad-CAM*
Built a multi-model deep learning pipeline for cryptocurrency trend prediction with Grad-CAM explainability. Bronze Research Excellence Award, IIT.

SalesForecaster, *Python, XGBoost, SHAP*
Time-series sales forecasting system with leakage-safe feature engineering and SHAP-based explainable AI for business-driven what-if analysis.

PUBLICATIONS

Image-Based Cryptocurrency Trend Prediction with Explainable Deep Learning ,
9th International Conference on Information Technology Research (ICITR 2024), IEEE
Bronze Research Excellence Award - Informatics Institute of Technology

EDUCATION

BEng Software Engineering - First Class Honours, *Informatics Institute of Technology (IIT), affiliated with the University of Westminster, UK* 2024

G.C.E. Advanced Level - Physical Science Stream, 2020
St. Sebastian's College, Moratuwa

EXTRA-CURRICULAR ACTIVITIES

Head Prefect, *St. Sebastian's College, Moratuwa* 2019

President - Batch of 2020, *St. Sebastian's College, Moratuwa*

Welfare Committee, *HCM - IFS R&D International* 2023

Semi-Finalist, *Hacktron - Game development hackathon* 2021

Public Relations, Rotaract Club | Member, Entrepreneurship Society, 2021 – 2023
Informatics Institute of Technology

REFERENCES

References available upon request.